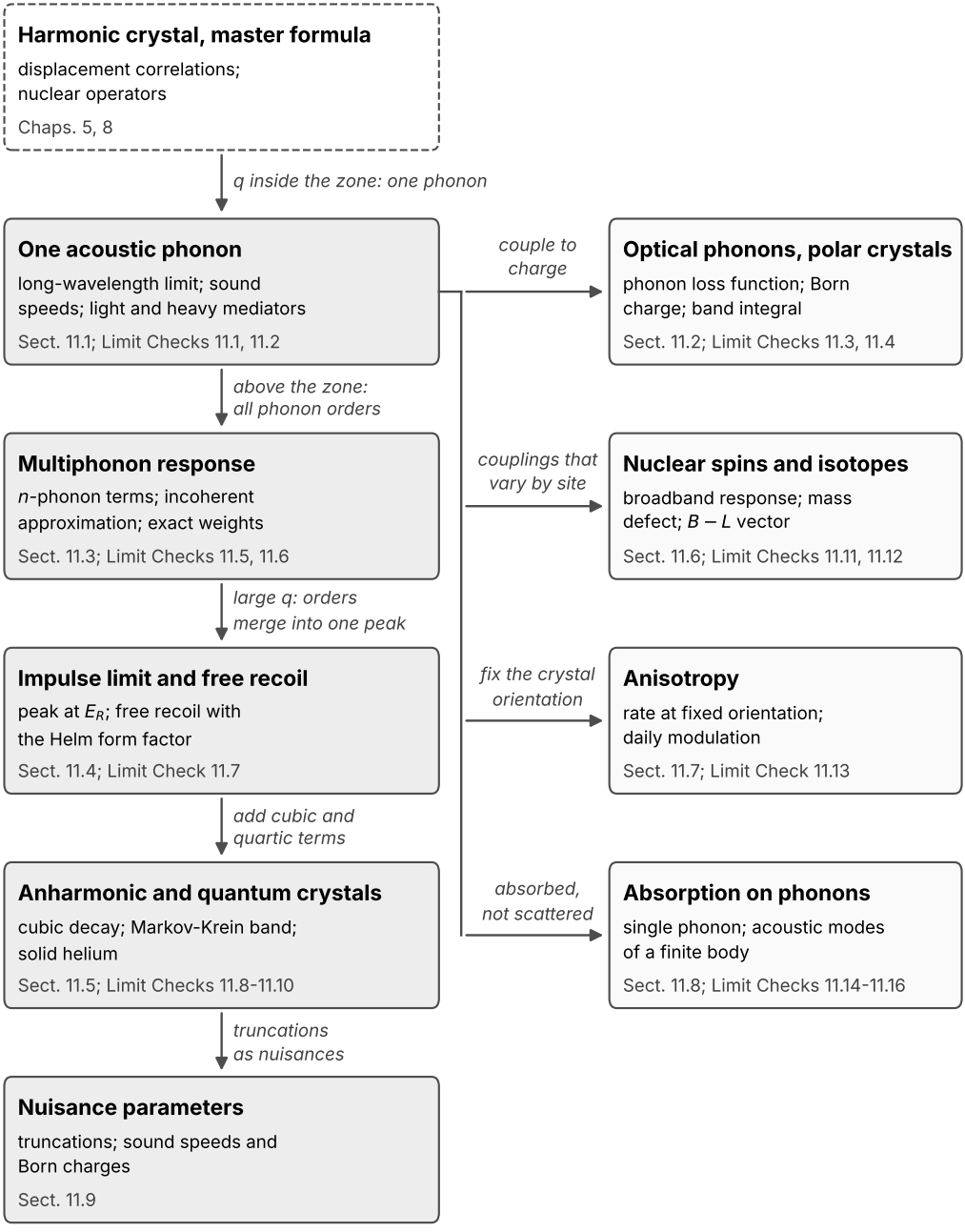
Harmonic crystal, master formula
displacement correlations;
nuclear operators
Chaps. 5, 8
q inside the zone: one phonon
One acoustic phonon
long-wavelength limit; sound
speeds; light and heavy mediators
Sect. 11.1; Limit Checks 11.1, 11.2
couple to
charge
Optical phonons, polar crystals
phonon loss function; Born
charge; band integral
Sect. 11.2; Limit Checks 11.3, 11.4
above the zone:
all phonon orders
Multiphonon response
n-phonon terms; incoherent
approximation; exact weights
Sect. 11.3; Limit Checks 11.5, 11.6
couplings that
vary by site
Nuclear spins and isotopes
broadband response; mass
defect; B − L vector
Sect. 11.6; Limit Checks 11.11, 11.12
large q: orders
merge into one peak
Impulse limit and free recoil
peak at E_R; free recoil with
the Helm form factor
Sect. 11.4; Limit Check 11.7
fix the crystal
orientation
Anisotropy
rate at fixed orientation;
daily modulation
Sect. 11.7; Limit Check 11.13
add cubic and
quartic terms
Anharmonic and quantum crystals
cubic decay; Markov-Krein band;
solid helium
Sect. 11.5; Limit Checks 11.8-11.10
absorbed,
not scattered
Absorption on phonons
single phonon; acoustic modes
of a finite body
Sect. 11.8; Limit Checks 11.14-11.16
truncations
as nuisances
Nuisance parameters
truncations; sound speeds and
Born charges
Sect. 11.9